Sri Lanka Institute of Information Technology

Web-Based Transport System(Like Uber)

Final Report

Year 2 semester 1 (2025) – SE2030

Group ID :- **2025-Y2-S1-MLB-B11G2-08**

Submission Date – 21.10.2025

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Introduction

With the hecticness of life in today's times, individuals need transport services that are convenient, trustworthy as well as punctual. The traditional method, like stopping taxis on the street, is fast becoming a relic of the past. Although platforms like Uber have revolutionized city mobility, their services are never available or accessible in smaller towns or developing areas.

Our project, "QuickMove Lanka," is an internet-based transport system to be utilized as a local ride-hailing platform. The platform allows passengers to book rides online through an online, lightweight application that subsequently books them with drivers in the vicinity and delivers a seamless ride experience.

Objectives

The primary goal of this project is to improve the efficiency and accessibility of transport services. We aim to provide a simple, easy-to-use web application that works across all devices with internet access, thereby enhancing customer satisfaction and operational efficiency.

Target Users or Stakeholders

The system is designed for several key stakeholders:

- Riders (Customers): People who need an easy-to-use system to book rides and see details like cost and arrival time.
- Drivers: Registered drivers who use the system to accept trip requests, navigate to customers, and track their earnings.
- IT Admins: Technical staff responsible for managing user accounts, monitoring system performance, and resolving issues.
- Our Management Team: The team that uses system data and reports to make business decisions and improve the service.

Scope and Limitations

The scope of this project is to develop a functional web-based application that handles the complete ride-booking lifecycle, including user registration, booking, driver matching, real-time tracking, and admin management.

The project also operates under the following limitations :

1. The system's performance is dependent on stable internet connectivity for both drivers and riders.
2. GPS accuracy may be a limitation in rural areas.
3. High user traffic could cause server load if not managed with scalable hosting.

Requirements

Functional Requirements

User Registration and Login

- User Account Creation: Allow customers, drivers, and admins to create accounts with personal details.
- Email/Phone Verification: Verify user identity via email or phone number.
- Secure Login: Provide secure authentication for all user types.
- Forgot Password: Enable password recovery via email or phone.
- Edit Profile: Allow users to update personal information.
- Logout: Provide option to securely log out.

Ride Booking System

- Search for Nearby Drivers: Display available drivers near the customer's location.
- Fare Estimation: Calculate and display estimated ride cost.
- Select Pickup and Drop-off Location: Allow users to specify ride locations.
- Book Ride Now: Enable immediate ride booking.
- Cancel Booking: Allow customers to cancel a booked ride.

Driver Ride Acceptance

- View Ride Requests: Display incoming ride requests to drivers.
- Accept/Reject Ride: Allow drivers to accept or decline requests.
- Start Ride: Enable drivers to mark ride start.
- End Ride: Allow drivers to mark ride completion.
- View Earnings: Display driver earnings per ride or period.
- Ride History: Provide access to past ride details

Real Time Ride Tracking

- Live-Map Tracking: Show real-time driver location on a map.
- Driver Arrival Notification: Notify customers when driver arrives.
- Route Optimization: Suggest optimal routes to drivers.
- Estimated Time of Arrival: Display ETA to customers.

Admin Dashboard for System Monitoring

- User Management: Enable admins to manage user accounts.
- View Active Rides: Monitor ongoing rides in real-time.
- Complaint Handling: Address user complaints efficiently.
- Reports and Analytics: Generate system usage and performance reports.
- Driver Verification: Verify driver credentials and documents.

Payment and Billing Integration

- Fare Calculation: Compute ride fare based on distance and time.
- Payment Gateway Integration: Support secure online payments.
- Cash Payment Option: Allow cash payments for rides.
- Invoice Generation: Provide detailed ride invoices.
- View Payment History: Display past payment records for users.

Non functional requirements

- Performance
  The system should be capable of handling multiple user requests simultaneously without performance degradation. All user actions, including ride requests and page transitions, should be processed quickly to ensure a smooth user experience.
- Security
  Security is critical, especially since the system handles user identities, location data, and payment information. Passwords must be stored using secure hashing algorithms. User roles (admin, driver, rider) must be protected using role-based access control.
- Usability
  The web application must be user-friendly with a clear, intuitive interface. Navigation should be consistent, and all forms should include proper validation and error messages.
- Scalability
  System should support future expansion (e.g., more users, new features like chat, promo codes).Easy to integrate with third-party services like maps and payment gateways.
- Reliability
  The application must ensure stable and consistent behavior, even in cases of unexpected user behavior or system failures. If a server crash occurs, it should not result in data loss or system corruption.

Constraints and Assumptions

Constraints

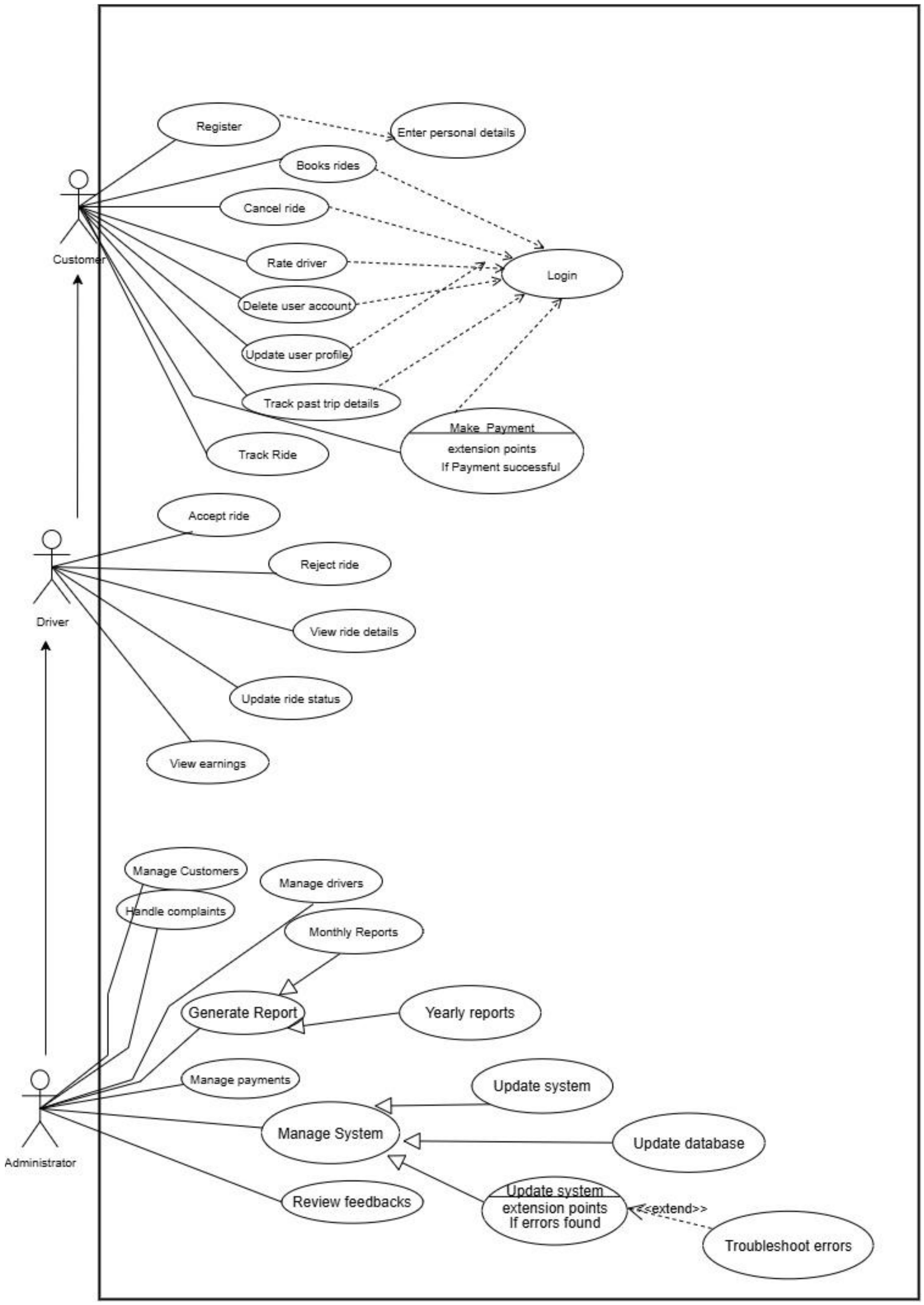
- The system must be developed as a web-based application; a native mobile app is not in scope.
- The web application must be responsive and support both mobile and desktop browsers.
- System performance is constrained by server load and may slow down if many users access it simultaneously.

Assumptions

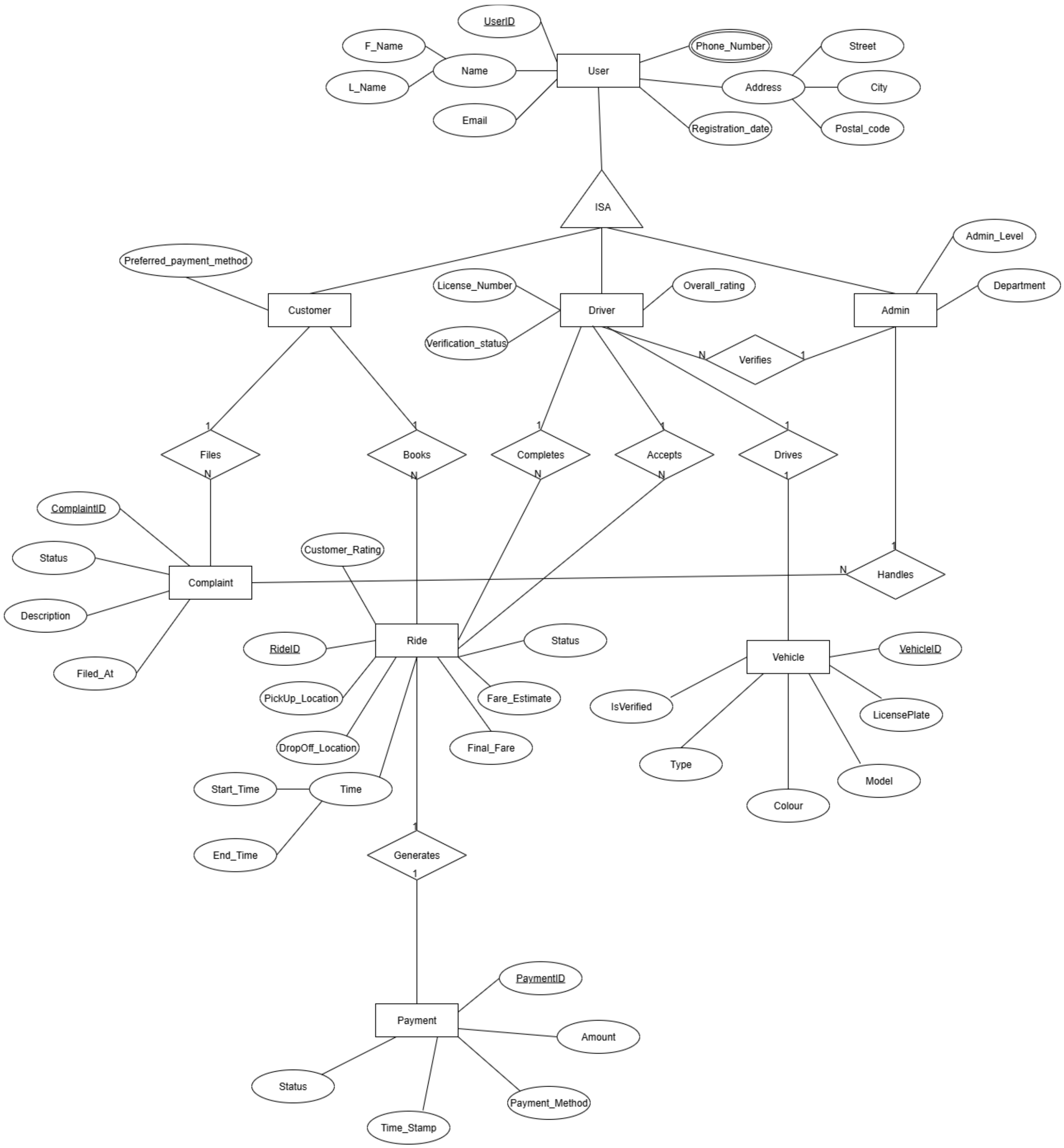
- It is assumed that both Riders and Drivers have a stable internet connection.
- We assume users will have their device's GPS enabled for location services.
- We assume users have the basic technical knowledge required to operate a web application.
- We assume that GPS location data may be inaccurate in areas with poor satellite reception.

Design

Use Case Diagram



ER diagram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Registration_date
ISA
Preferred_payment_method
Customer
License_Number
Verification_status
Driver
Overall_rating
Admin
Admin_Level
Department
Verifies
Files
Books
Completes
Accepts
Drives
Handles
ComplaintID
Status
Description
Filed_At
Complaint
Customer_Rating
Ride
RideID
PickUp_Location
DropOff_Location
Start_Time
Time
End_Time
Fare_Estimate
Final_Fare
Generates
Vehicle
VehicleID
IsVerified
LicensePlate
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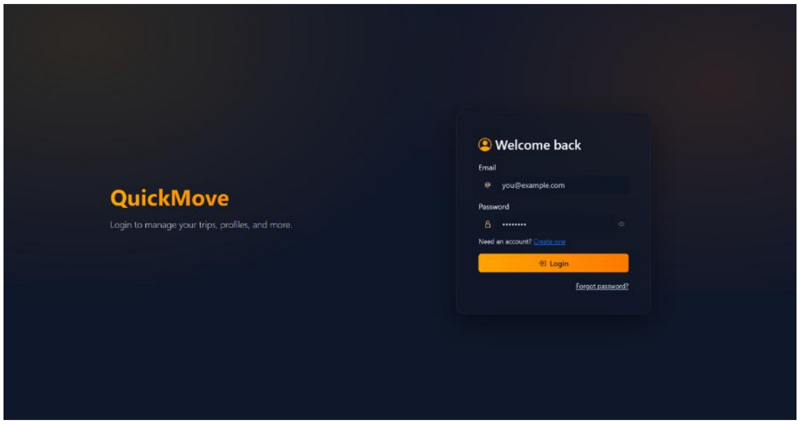
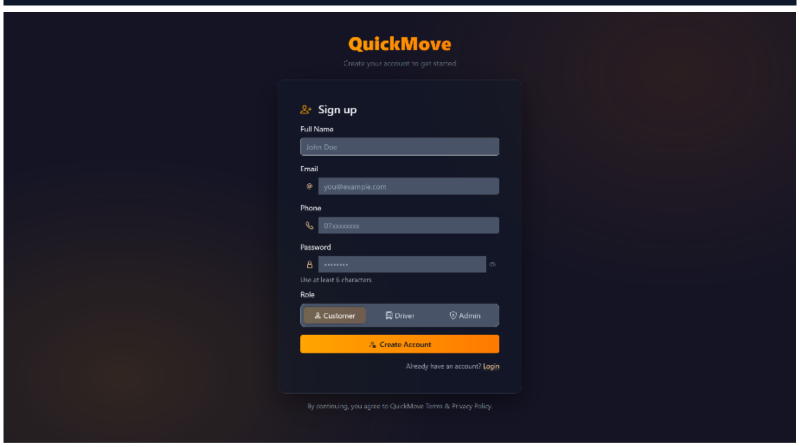
UI Sketches/ Screenshots and Descriptions

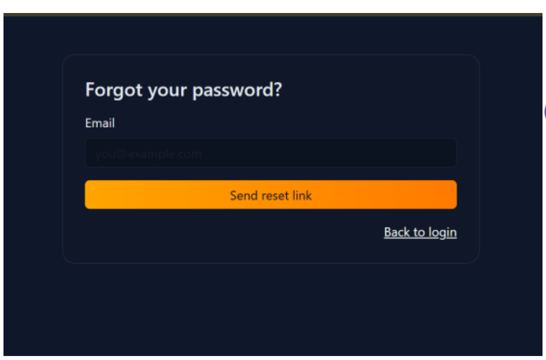
User Management UI

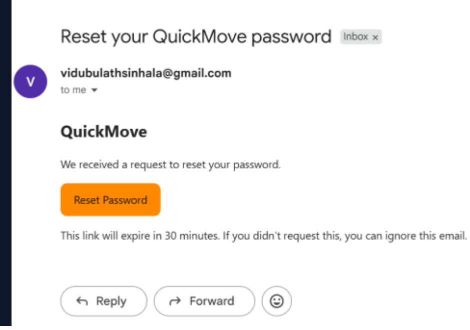
- Register & Sign-in

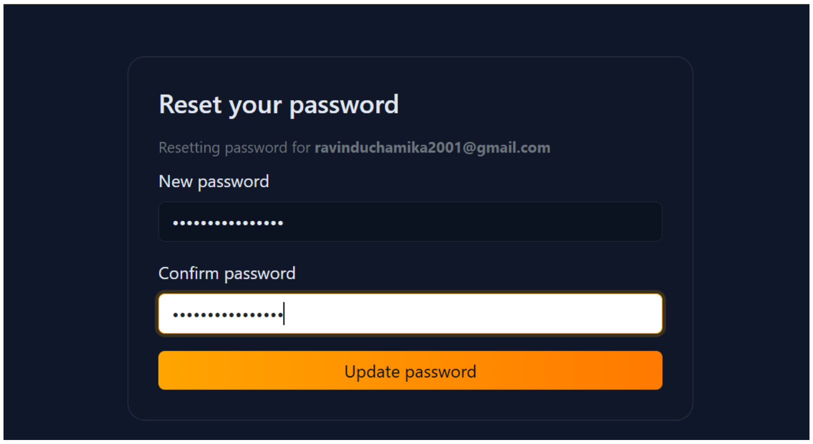




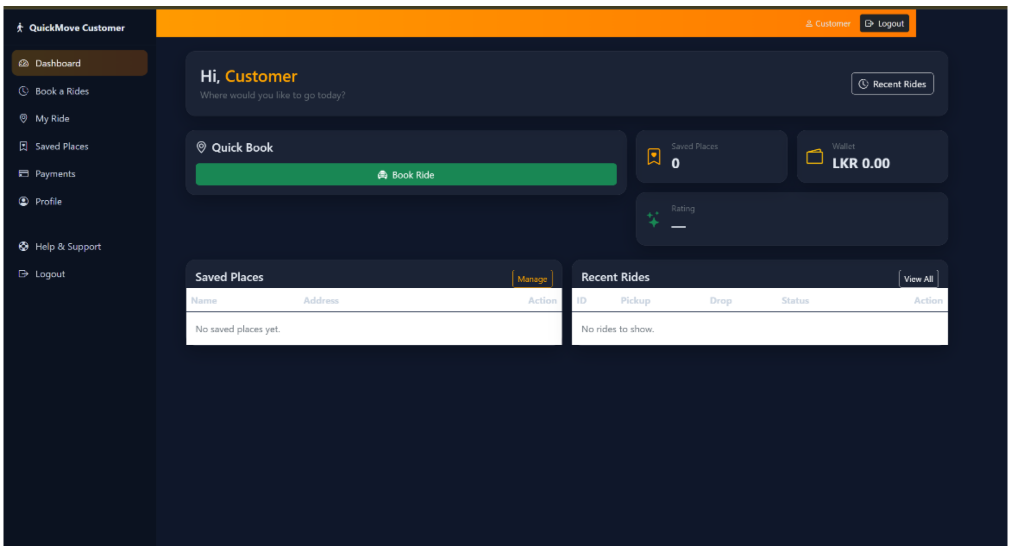
- Forgot Password with Email verification

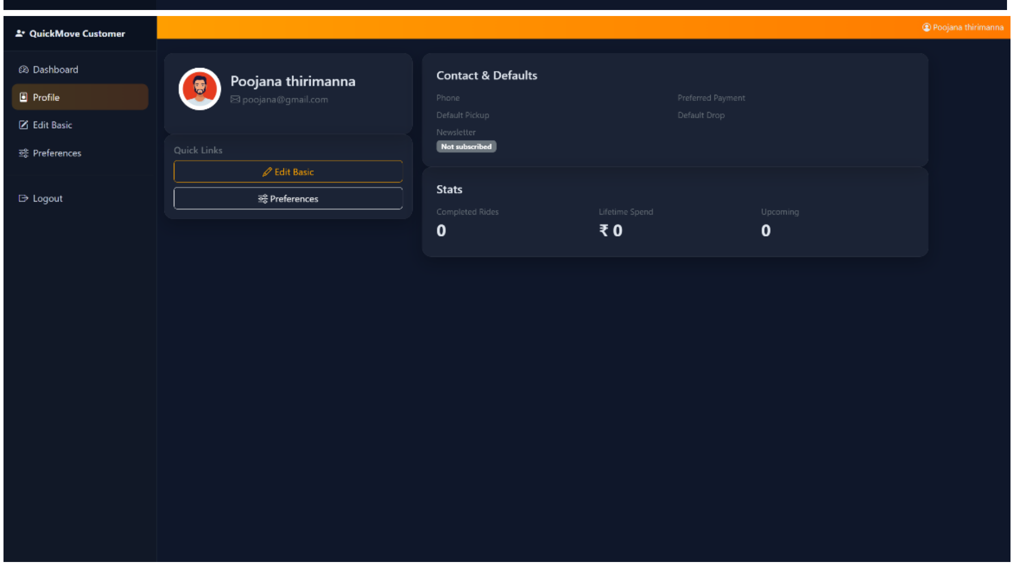


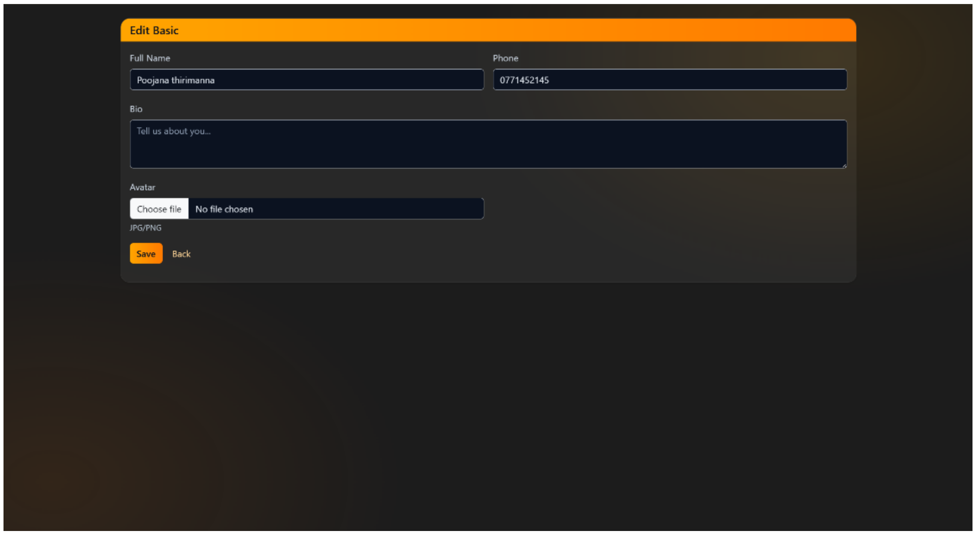




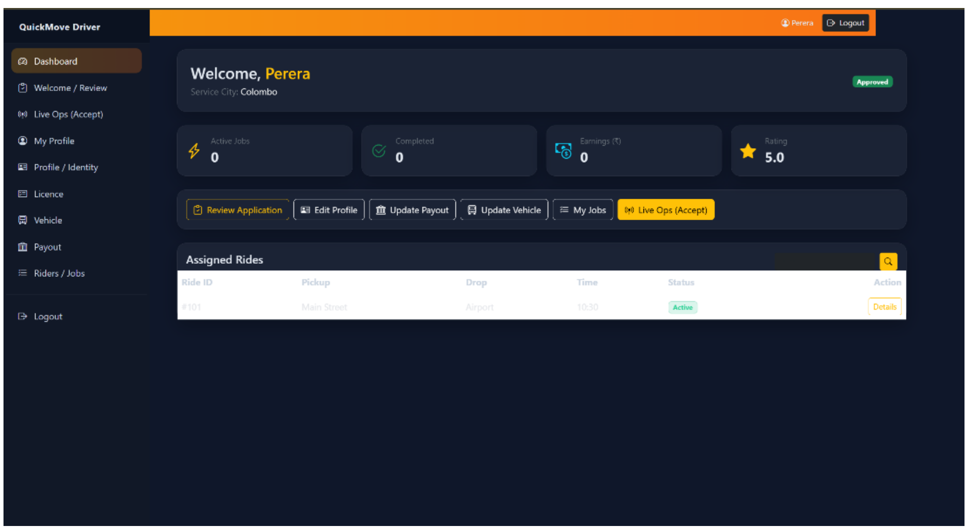
- Customer Dashboard & Customer Profile

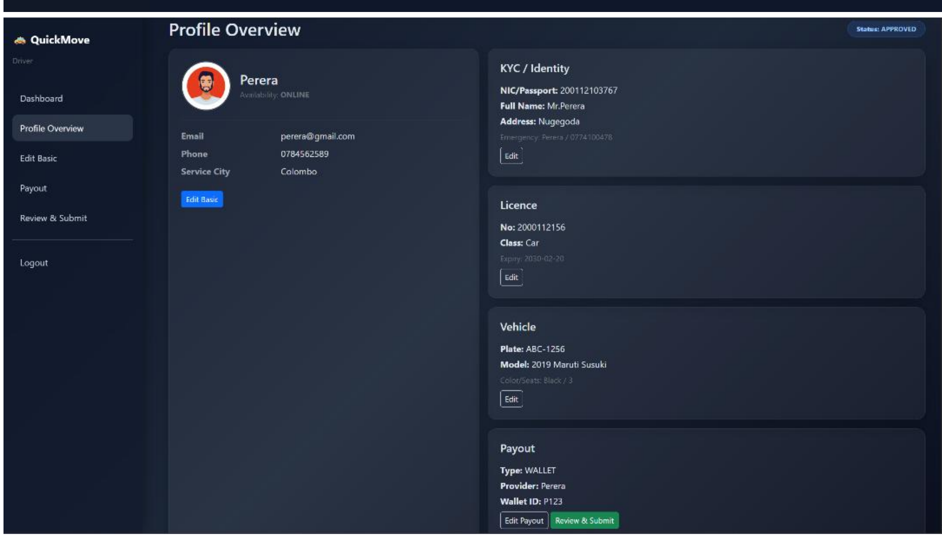


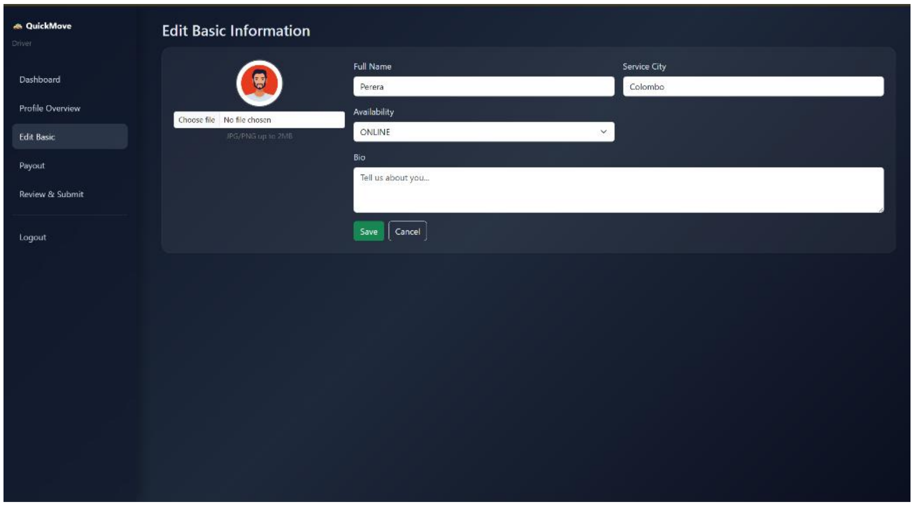




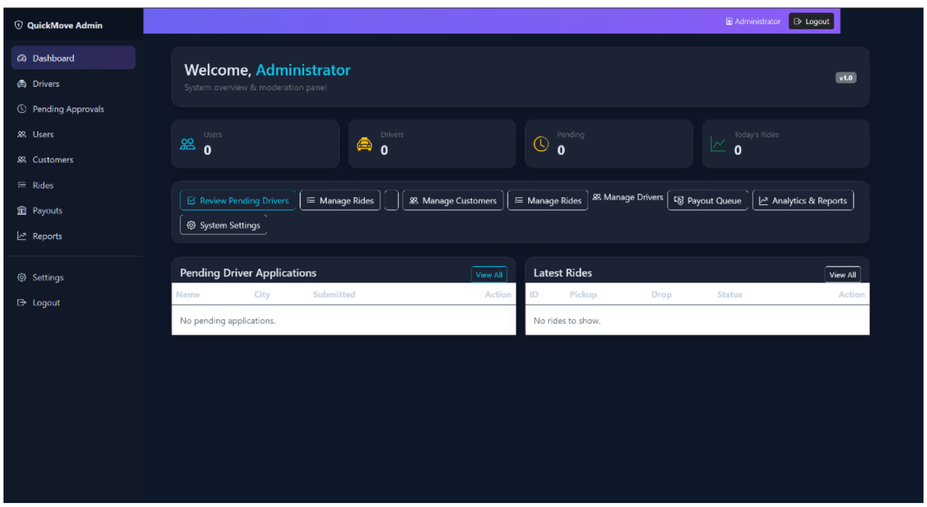
- Driver Dashboard & Driver Profile

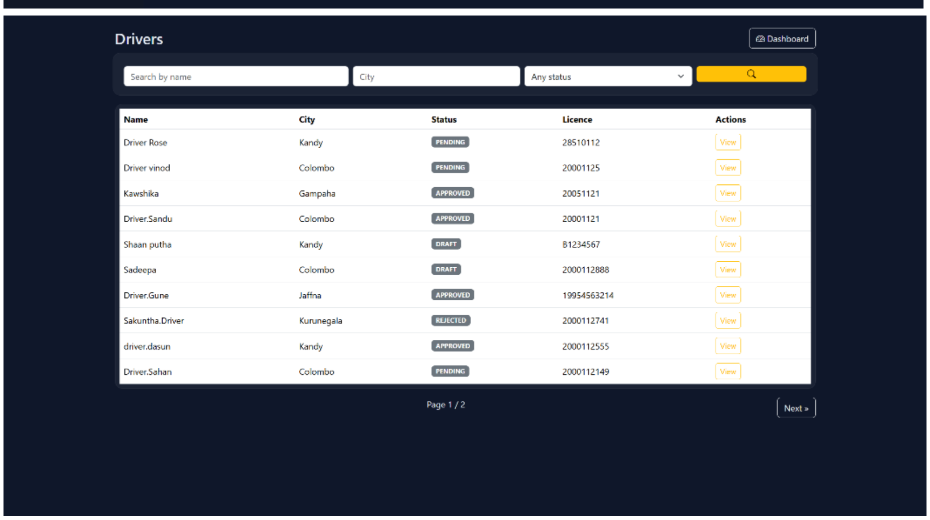


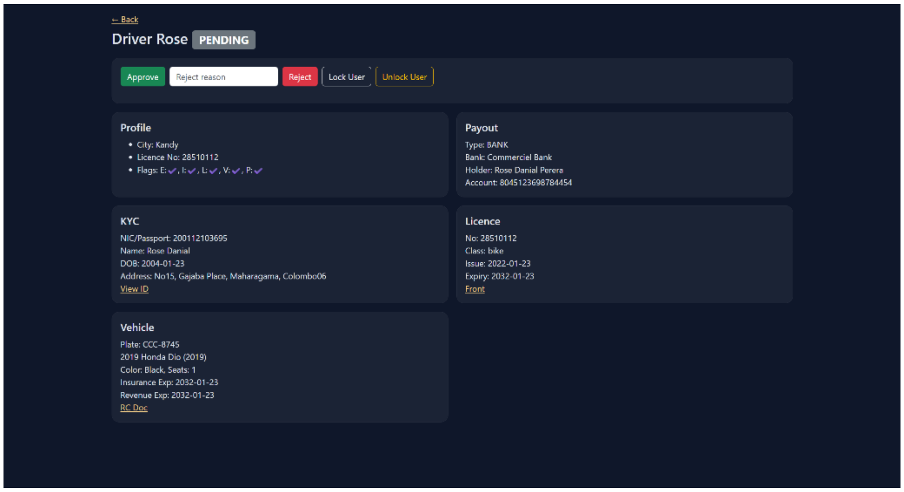




- Admin Dashboard



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Driver Rose
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Holder: Rose Danial Perera
Account: 8045123698784454
KYC
NIC/Passport: 200112103695
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Revenue Exp: 2032-01-23
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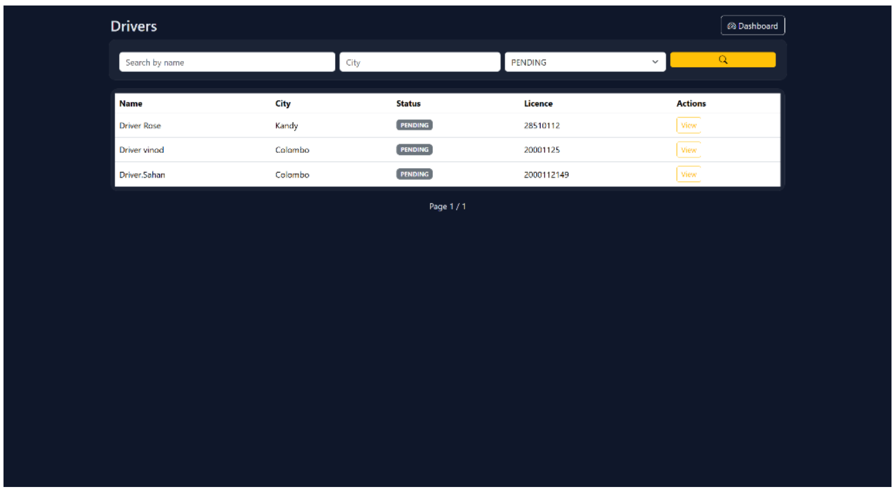
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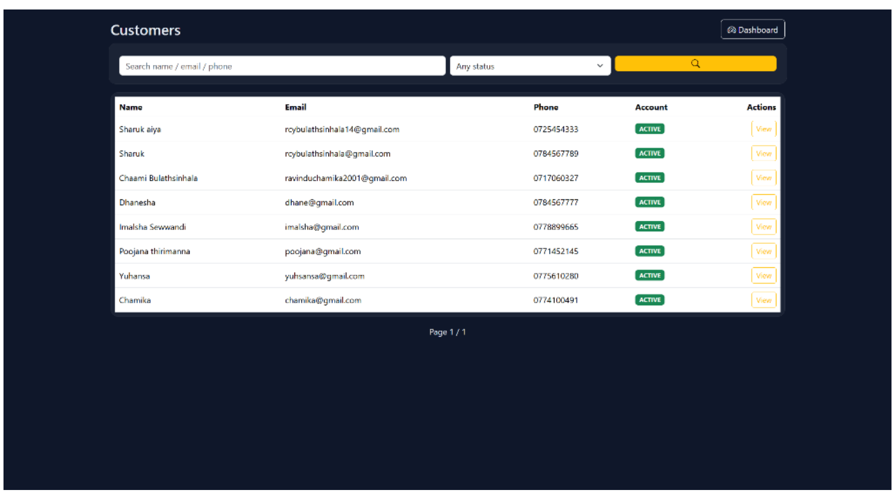
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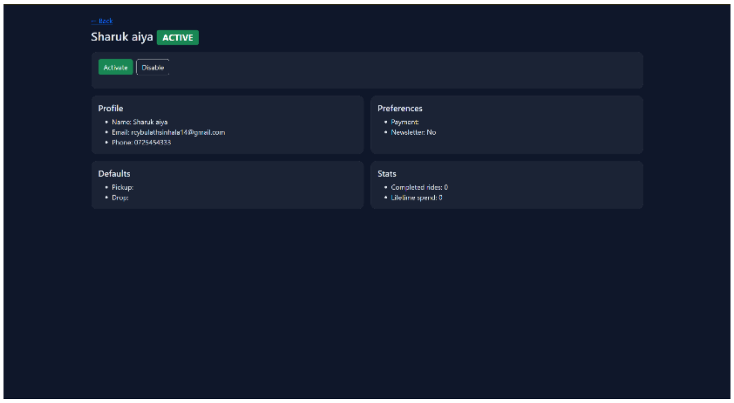
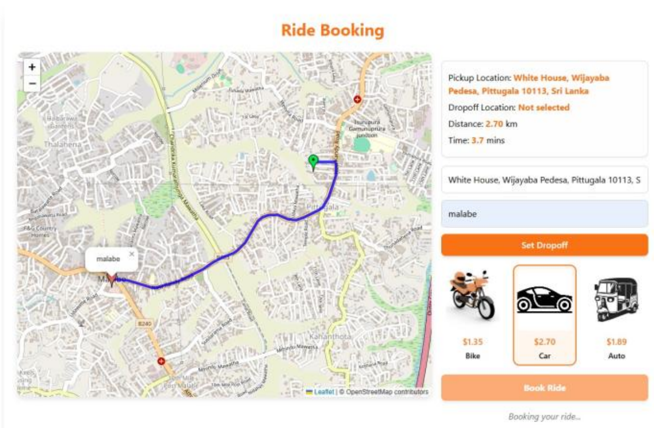
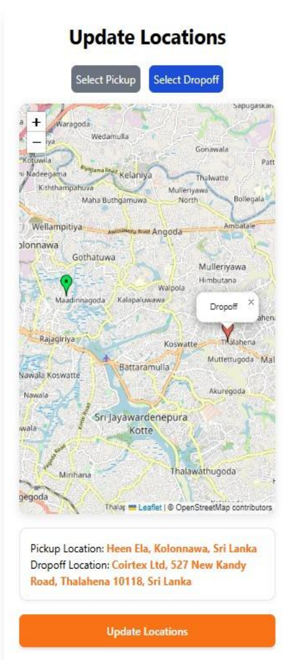
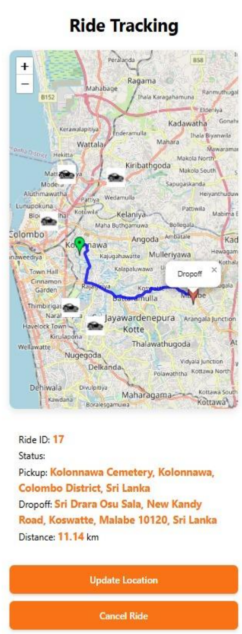
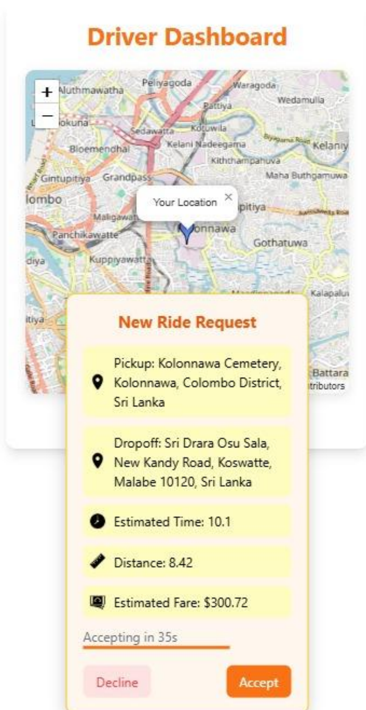


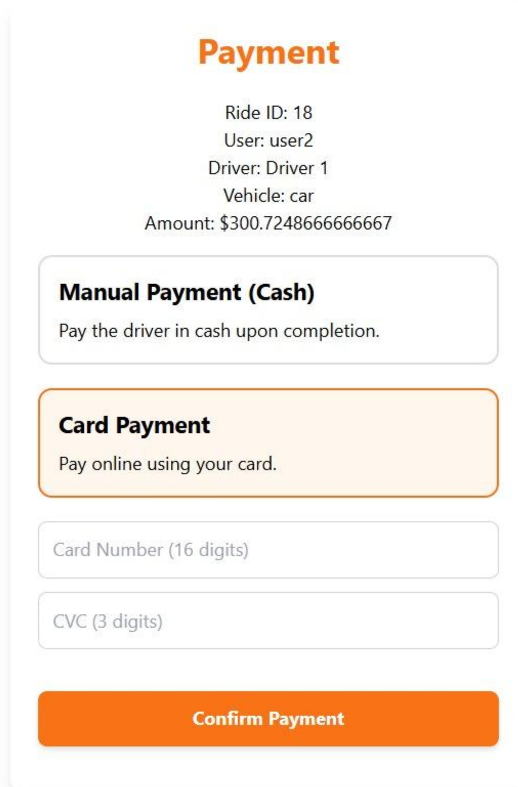
Ride Booking, Driver Ride Acceptance and payment UI



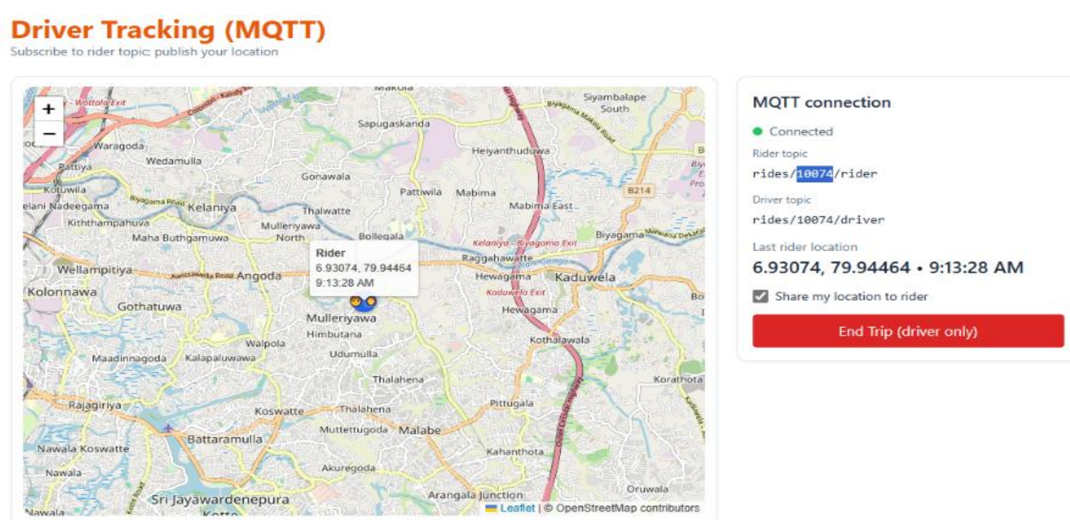


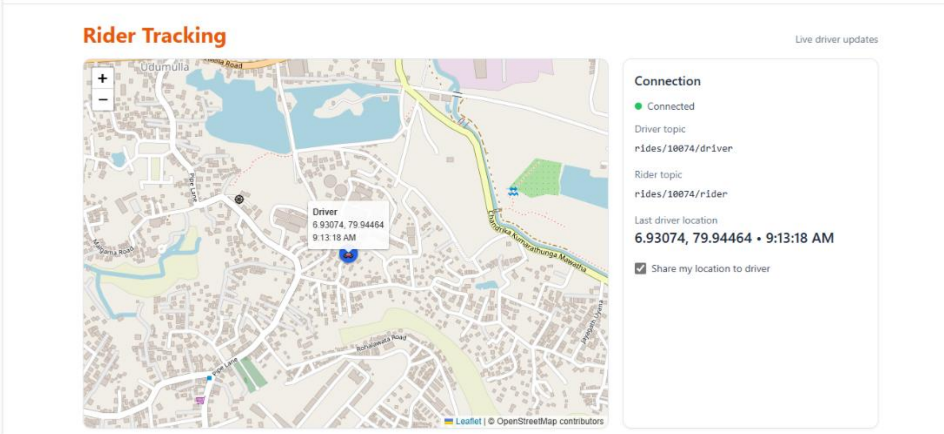




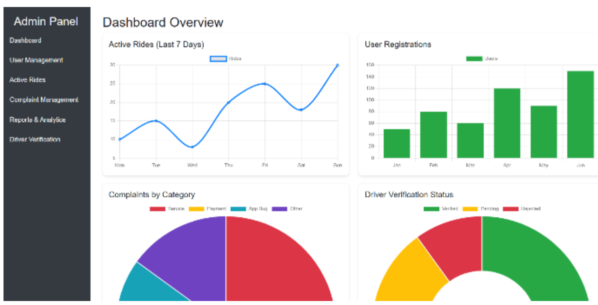


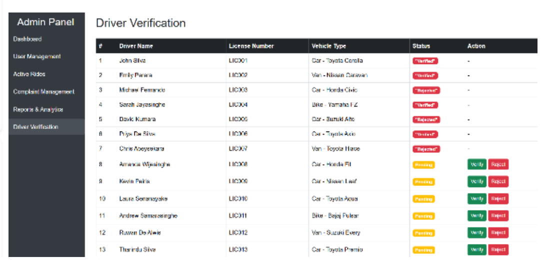
Ride Tracking UI



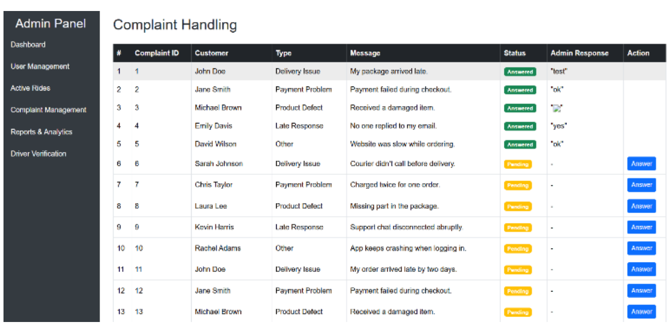


Admin Dashbord UI

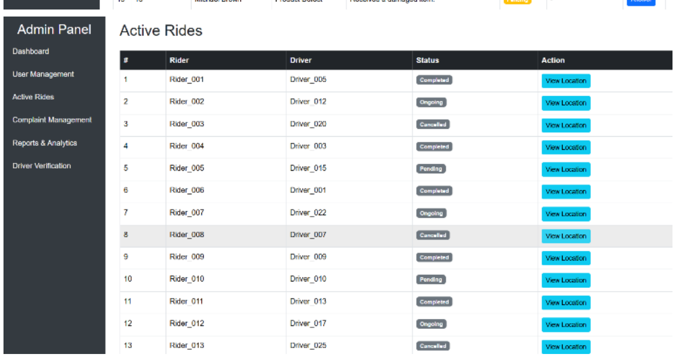




#	Driver Name	License Number	Vehicle Type	Status	Action
1	John Silva	LIC001	Car - Toyota Corolla	"Verified"	-
2	Emily Perera	LIC002	Van - Nissan Caravan	"Verified"	-
3	Michael Fernando	LIC003	Car - Honda Civic	"Rejected"	-
4	Sarah Jayasinghe	LIC004	Bike - Yamaha FZ	"Verified"	-
5	David Kumara	LIC005	Car - Suzuki Alto	"Rejected"	-
6	Priya De Silva	LIC006	Car - Toyota Axio	"Verified"	-
7	Chris Abeysekara	LIC007	Van - Toyota Hiace	"Rejected"	-
8	Amanda Wijesinghe	LIC008	Car - Honda Fit	Pending	Verify Reject
9	Kevin Peiris	LIC009	Car - Nissan Leaf	Pending	Verify Reject
10	Laura Senanayake	LIC010	Car - Toyota Aqua	Pending	Verify Reject
11	Andrew Samarasinghe	LIC011	Bike - Bajaj Pulsar	Pending	Verify Reject
12	Ruwan De Alwis	LIC012	Van - Suzuki Every	Pending	Verify Reject
13	Tharindu Silva	LIC013	Car - Toyota Premio	Pending	Verify Reject



#	Complaint ID	Customer	Type	Message	Status	Admin Response	Action
1	1	John Doe	Delivery Issue	My package arrived late.	Answered	"test"	
2	2	Jane Smith	Payment Problem	Payment failed during checkout.	Answered	"ok"	
3	3	Michael Brown	Product Defect	Received a damaged item.	Answered	""	
4	4	Emily Davis	Late Response	No one replied to my email.	Answered	"yes"	
5	5	David Wilson	Other	Website was slow while ordering.	Answered	"ok"	
6	6	Sarah Johnson	Delivery Issue	Courier didn't call before delivery.	Pending	-	Answer
7	7	Chris Taylor	Payment Problem	Charged twice for one order.	Pending	-	Answer
8	8	Laura Lee	Product Defect	Missing part in the package.	Pending	-	Answer
9	9	Kevin Harris	Late Response	Support chat disconnected abruptly.	Pending	-	Answer
10	10	Rachel Adams	Other	App keeps crashing when logging in.	Pending	-	Answer
11	11	John Doe	Delivery Issue	My order arrived late by two days.	Pending	-	Answer
12	12	Jane Smith	Payment Problem	Payment failed during checkout.	Pending	-	Answer
13	13	Michael Brown	Product Defect	Received a damaged item.	Pending	-	Answer



#	Rider	Driver	Status	Action
1	Rider_001	Driver_005	Completed	View Location
2	Rider_002	Driver_012	Ongoing	View Location
3	Rider_003	Driver_020	Cancelled	View Location
4	Rider_004	Driver_003	Completed	View Location
5	Rider_005	Driver_015	Pending	View Location
6	Rider_006	Driver_001	Completed	View Location
7	Rider_007	Driver_022	Ongoing	View Location
8	Rider_008	Driver_007	Cancelled	View Location
9	Rider_009	Driver_009	Completed	View Location
10	Rider_010	Driver_010	Pending	View Location
11	Rider_011	Driver_013	Completed	View Location
12	Rider_012	Driver_017	Ongoing	View Location
13	Rider_013	Driver_025	Cancelled	View Location

Implementation

Overview

The implementation of QuickMove Lanka transformed the project design into a fully functional ride-hailing system using Spring Boot, following the MVC architecture with a Repository layer. The system provides the following core functionalities.

- ♦ **User Registration, Login & Authentication**
- ♦ **Driver Ride Acceptance**
- ♦ **Real-Time Ride Tracking**
- ♦ **Ride Booking System**
- ♦ **Payment and Billing**
- ♦ **Admin Dashboard for system monitoring**

The system architecture ensures a clear separation of concerns:

- **Model Layer:** Entity classes representing database tables (User, Driver, Ride, Payment) using **Spring Data JPA**.
- **Controller Layer:** Handles HTTP requests, calling services, and returning views or JSON responses.
- **Repository Layer:** Interfaces extending JpaRepository for database operations.
- **Service Layer:** Implements business logic and mediates between controllers and repositories.
- **View Layer:** HTML, CSS, and JavaScript templates providing an intuitive and responsive user interface.

Tools and Technologies Used

Technology / Tool	**Purpose / Description**
Java	Core programming language
Spring Boot	Backend framework with MVC support
Spring Data JPA / Hibernate	ORM and repository-based database access
Microsoft SQL Server	Relational database for users, drivers, rides, and payments
Spring Security & JWT	Secure authentication and authorization
STOMP over SockJS	Real-time messaging for driver ride acceptance
MQTT over WebSocket	Real-time ride tracking updates
HTML, CSS, JavaScript	Frontend UI development
IntelliJ IDEA	IDE for development and debugging
Postman	API testing
GitHub	Version control and collaboration

Core Functionality Implementation

Function	Implementation Highlights
User Registration, Login & Authentication	Implemented via UserController, UserService, and UserRepository. Passwords are encrypted using BCryptPasswordEncoder. Login uses Spring Security and JWT for session management.
Driver Ride Acceptance	Drivers receive ride requests and status updates in real-time using STOMP over SocketJS. Accepted rides are updated in RideRepository and passengers receive instant notifications.
Real Time Ride Tracking	Driver GPS coordinates are published via MQTT over WebSocket and displayed live on the passenger interface.
Ride Booking System	Users submit ride requests through RideController. RideService handles assignments and updates ride status in the database via RideRepository.
Payment and Billing	PaymentService calculates fares, updates PaymentRepository, and generates invoices.
Admin Dashboard	Aggregates system data from repositories to display analytics and manage users/drivers. Charts and tables provide real-time insights.

Project Management

Agile Approach and Sprint Summary

We adopted an Agile Methodology, organizing our work into four distinct sprints. This allowed us to iteratively develop features, test them and adapt to changes.

- Sprint 1 Summary(Week 4 - Week 6)
  Planned:- To create and authenticate customer accounts (PBI-01), implement the ride-booking workflow (PBI-02), and build the initial operations dashboard (PBI-03).
  Completed:- The registration and login feature was completed. The ride-booking workflow, including Google Maps API integration, was also completed. The dashboard UI was implemented, and testing was carried out for these flows.

- Sprint 2 Summary(Week 7 - Week 9)
  Planned:- To add a booking search feature for customer support (PBI - 04), implement driver account management (PBI -05), and add driver rating and complaint monitoring (PBI - 06).
  Completed:- A search facility for booking was successfully deployed. The driver administration portal(add/edit/remove) was also deployed. Driver rating and complaint logging features are working

- Sprint 3 Summary(Week 10 - Week 12)
  Planned:- To implement a full complaint logging and tracking system (PBI - 07) and create admin controls for system rules like fare rates (PBI - 08).
  Completed:- A complete complaint ticketing system with status tracking was put in place. The admin interface for setting fare rates is functional.

- Sprint 4 Summary(Week 13 – Week 14)
  Planned:- To implement driver compliance monitoring using map data (PBI - 09) and a system for collecting customer feedback (PBI - 10).
  Completed:- The compliance dashboard was created. Customer feedback forms were designed and linked to the backend, along with a simple feedback analysis dashboard.

Task Distribution among Team Members

Each of the six major functions of the project was assigned to a specific team member, who took responsibility for its design and implementation.

Student ID	Name	Assigned Major Function
IT24103019	Illeperuma H.P	Driver Ride Acceptance
IT24103028	De Silva G.Y.S	User Management
IT24103113	Wikejoon W.H.M.P.V.P	Real Time Ride Tracking
IT24103106	Ranathunga V.V.S.B.U	Payment and Billing
IT24103052	Sukirthan C	Ride Booking System
IT24103030	Arampath A.M.G.C.L	Admin Dashboard

Development Phases or Timeline

The project followed the high-level timeline established in our initial proposal. The key milestones were mapped from week 3 to week 14.

- Week 3 : Proposal Presentation and Report Submission
- Week 4 : Finalize System Design
- Week 5-6 : UI/UX Design
- Week 7-9 : Core Backend and Frontend Development
- Week 10 : Progress Presentation and evaluation
- Week 11-12 : Bug fixing and System Improvements
- Week 13 : Final testing and documentation
- Week 14 : Final Presentation and System Demonstration

Conclusion and Future Work

Conclusion

In conclusion , our project “QuickMove Lanka” successfully achieved its main goal of building a simple, reliable, and efficient web-based transport system for local users. The system allows passengers to book rides easily, track drivers in real time, make secure payments, and gives admins full control through a management dashboard. By using Spring Boot and the MVC architecture, we were able to create a well-structured, scalable, and easy-to-maintain system. Overall, this project helped us understand how technology can be used to make everyday transportation more convenient, especially in areas where services like Uber are not easily available.

Future work

Although the current system meets our project objectives, there are still many ways it can be improved in the future. Some of the features we hope to add include:

- Developing a mobile app version for Android and iOS users.
- Using AI-based ride matching to connect drivers and passengers faster.
- Adding in-app chat and an SOS feature for better safety and communication.
- Introducing dynamic pricing that adjusts fares depending on demand and time.
- Supporting multiple languages such as Sinhala and Tamil for easier access.
- Connecting with public transport systems to help users plan complete journeys.

With these improvements, “QuickMove Lanka ”could grow into a smarter and more inclusive platform that better serves the needs of people across Sri Lanka.

Individual Contribution, Teamwork and Lessons learned

Name / ID	Role & Responsibilities	Challenges Faced	How Challenges Were Overcome	Key Lessons Learned
IT24103019	Developed the Driver Ride Acceptance module to allow drivers to view and accept rides efficiently.	Faced issues with synchronization between driver status and ride requests.	Implemented real-time database updates and optimized backend communication to ensure accuracy.	Learned about asynchronous request handling and data consistency in multi-user systems.
IT24103028	Designed and implemented the User Registration and Login system with authentication and input validation.	Faced difficulties integrating secure authentication.	Solved by using encryption and implementing input sanitization.	Learned secure user management and best practices in authentication.
IT24103113	Built the Real-time Tracking module using GPS to display driver and passenger locations.	Encountered issues with map API integration and frequent location update delays.	Resolved by optimizing the API polling rate and implementing socket-based communication.	Learned API optimization, location-based services, and real-time data handling.
IT24103106	Developed the Payment and Billing Integration module to manage transactions between users and drivers.	Faced challenges linking ride data with transaction records .	Overcame by applying transactional database logic.	Learned financial data handling .
IT24103052	**Created the** Ride Booking System allowing users **to request and schedule rides easily.**	**Encountered booking conflicts when multiple users requested rides simultaneously.**	**Fixed by adding a** queue management system and **database locking mechanisms.**	**Learned about** concurrency control and proper backend synchronization**.**
IT24103030	Designed the Admin Dashboard for system monitoring and management of users, rides, and payments.	Faced challenges in summarizing large datasets for admin analytics.	Solved by using data aggregation queries and pagination to improve performance.	Learned how to build data-driven dashboards and manage system performance.

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Appendix

GitHub Repository Link

The GitHub repository for this project serves as the central platform for version control, collaboration, and source code management throughout the development of the **Web-Based Transport Management System**. It contains all the project files, including the source code, documentation, and resources used during implementation. Using GitHub allows the team to work collaboratively, track changes efficiently, and maintain an organized development workflow. The repository also showcases the project's structure, coding practices, and progress history, reflecting the application of software engineering principles in a real-world context. Complete project can be accessed through the following link.

GitHub Repository Link:
https://github.com/IT24103052/web_based_transport_system

Note: The GitHub repository for this project contains all the source code, documentation, and related resources used during the development of the **Web-Based Transport Management System**.. It was used for version control, collaboration, and tracking progress throughout the project. However, this repository is currently set to private to maintain the confidentiality of the project files and codebase. Access can be granted upon request for evaluation or academic purposes.